

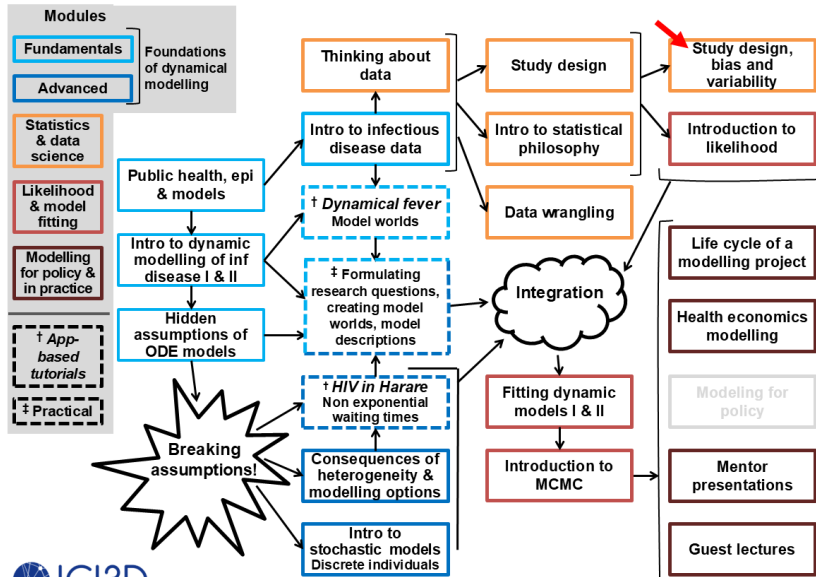
Recap

Study design, bias, variability

Clinic on the Meaningful Modelling of Epidemiological Data
MMED 2026

Jonathan Dushoff

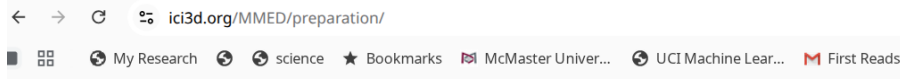
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Goals

- ▶ Go through some basic ideas from the preparatory study-design lectures
 - ▶ These are highly recommended
- ▶ Discuss the role of simulation in study design
- ▶ Introduce study-design coding session

Study design and data materials



have one, you can create a free account.

5. Thinking about data and study design: V

Relevant epidemiological data are crucial to the modelling of infe studies that generate them, there are four online lectures to wat

- Thinking about Data I ([one video](#)) - Slides
- Thinking about Data II ([part 1](#), [part 2](#), [part 3](#)) - Slides
- Study Design and Analysis in Epidemiology I ([part 1](#), [part 2](#), [part 3](#)) - Slides
- Study Design and Analysis in Epidemiology II ([part 1](#), [part 2](#), [part 3](#)) - Slides

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Outline

Study approaches

- Experimental design

- Observational data

Coding intro

- Whooping cough

Study

estimate =

truth

+

bias

+

random
noise



(accuracy)

(precision)

R. Kassanjee

Outline

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Example: weight-loss drug

- ▶ I enroll 60 people in an experiment to test a new weight-loss drug
- ▶ They all take the drug for six months, and I weigh them before and after
- ▶ Do you have any concerns?

Outline

Study approaches

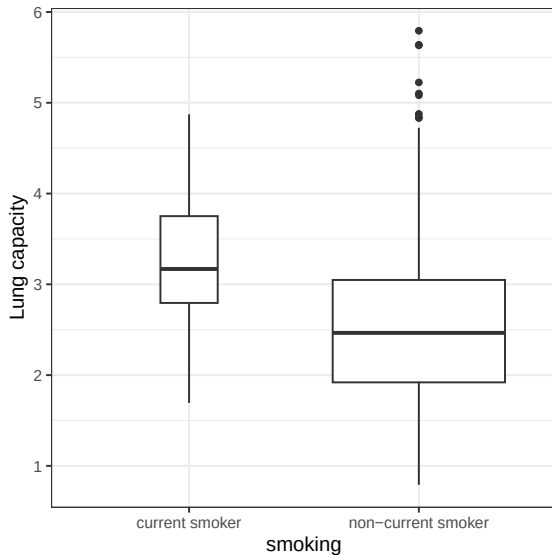
Experimental design

Observational data

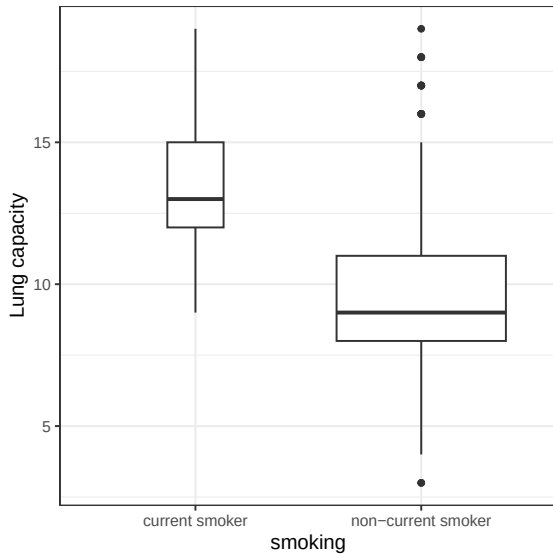
Coding intro

Whooping cough

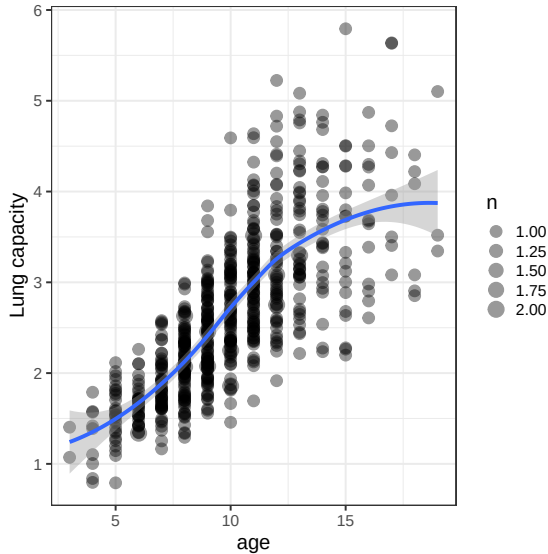
Example: smoking is good for you



Smoking data



Smoking data



Why do observational studies?

- ▶ Time
- ▶ Money
- ▶ Ethics

Confounders

- ▶ You can never remove all the bias from an observational study
 - ▶ * There will always be some difference between your groups that's out of your control
- ▶ But you can use statistical techniques to take some of the differences into account

Confounders again

- ▶ Confounders don't bias a properly randomized controlled study
- ▶ But we still often want to take them into account
 - ▶ * Otherwise, they can increase variability
- ▶ Best practice is to plan in advance to control for confounders expected to be important

Variability

- ▶ The best way to reduce variability is ...
 - ▶ * Have a bigger sample size
- ▶ Unfortunately, controlling for confounders can also increase variability
 - ▶ A type of overfitting

Summary

- ▶ Seek truth by asking good questions
- ▶ Reduce variability by
 - ▶ increasing sample size
 - ▶ careful planning (avoid overfitting)
- ▶ Reduce bias by
 - ▶ control and randomization when possible
 - ▶ accounting for key confounders otherwise

estimate =

$$\begin{array}{ccccc} \text{truth} & + & \text{bias} & + & \text{random} \\ & & & & \text{noise} \\ \img alt="Icon of a person with a lightbulb above their head, representing truth." data-bbox="651 478 693 538" & & \img alt="Icon of a blue monster with horns and a red tongue, representing bias." data-bbox="738 478 780 538" & & \img alt="Icon of two dice, representing random noise." data-bbox="831 491 873 538 \\ & & \text{(accuracy)} & & \text{(precision)} \end{array}$$

Outline

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Whooping cough

Planning

- ▶ Always a good idea to test your study approach in the computer
- ▶ Do you expect your study to work?
- ▶ Could you design it more efficiently

Validity

- ▶ Test by simulated experiments
- ▶ Bottom line: in the counter-factual case where nothing is going on, you want to find nothing clear (no statistical significance) $1 - \alpha$ of the time (typically 95%)
- ▶ Better: true effect should be inside the confidence interval 95% of the time
- ▶ Best: the P-value histogram for the true effect should be flat

Power

- ▶ Also test by simulated experiments
- ▶ If your hypothesis is right, what proportion of the time will your experiment work?
 - ▶ Show a clear result with the right sign
 - ▶ Or, show a clear “important” result with the right sign
- ▶ The first requires hypothesizing an effect size (or range of them)
- ▶ The second requires setting a second effect size as well!

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Whooping cough

- ▶ A bacterial disease which is re-emerging due to vaccine coverage problems and immune waning
- ▶ Cebile (and other people) want to know if adding a maternal vaccine to the schedule will help protect small children
- ▶ Could this be tested with a trial?



E D C T P



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Title: Recap

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